



FIRST SEMESTER 2026-2027
(COURSE HANDOUT PART II)

Date: 4/08/2026

In addition to Part I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No : CS G553
Course Title : Reconfigurable Computing
Instructors/in-charge : Meetha V Shenoy
Instructors : Jaya Sharma

- 1. Course Description:** Overview of Programmable Logics. FPGA fabric architectures. Logic Elements and Switch Networks. Design and Synthesis of Combinational and Sequential Elements. Placement and Routing. Pipelining and other Design Methodologies. Fine-grained and Coarse-Grained FPGAs. Static and Dynamic Reconfiguration. Partitioning. Hardware/Software Partitioning and Partial Evaluation; Systolic Architectures. Users perspective of reconfigurability – implementation and use cases on modern FPGAs.
- 2. Scope and Objective:** Reconfigurable (adaptive) computing is a novel yet important research field investigating the capability of hardware to adapt to changing computational requirements such as emerging standards, late design changes, and even to changing processing requirements arising at run-time. Reconfigurable computing thus benefits from a) the programmability of software similar to the Von Neumann computer and b) the speed and efficiency of parallel hardware execution. The purpose of the course reconfigurable computing is to instruct students about the possibilities and rapidly growing interest in adaptive hardware and corresponding design techniques by providing them the necessary knowledge for understanding and designing reconfigurable hardware systems and studying applications benefiting from dynamic hardware reconfiguration.

3. Course outcomes:

- CO1:** Identify the role and advantages of reconfigurable computing compared to other computing paradigms.
- CO2:** Understand FPGA design flow, placement, routing, and configuration aspects in reconfigurable systems.
- CO3:** Understand reconfiguration techniques such as partial and run-time reconfiguration in adaptive hardware systems.
- CO4:** Apply Verilog to model and implement digital designs on FPGA platforms.

4. Text Book

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1. Wolf Wayne, *FPGA Based System Design*, Pearson Edu, 2004.

5. Reference Book

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1. Scott Hauck, André DeHon, *Reconfigurable Computing - The Theory and Practice of FPGA Based Computation*, The Morgan Kaufmann Series in Systems on Silicon, 2007.
2. C Bobda, *Introduction to Reconfigurable Computing: Architectures, Algorithms, and Applications*, Springer, 2007.
3. R Vaidyanathan, Trahan Jerry, *Dynamic Reconfiguration: Architectures and Algorithms*, L, Kluwer Academic, 2003.
4. Uwe Meyer-Baese, *DSP with FPGAs*, Springer-Verlag, 2003.

6. Method of conduct of the course

Three lecture hours and 2 hours of lab per week.

7. Course Plan

Week	Module	Lecture No.	Learning Objectives	Topics to be covered
1	1	1-2	Introduction to reconfigurable computing	Introduction application and comparison - General Purpose Computing - Domain Specific Computing - Application Specific Computing - Reconfigurable Computing
		3	VLSI Technology	Wires, Registers and RAM - Wires and vias - Gate delay vs. wire delay - Registers and RAM
2	2	4-6	Reconfigurable Computing Hardware	Programmable logic, an overview of PLA, PAL, SPLD and CPLD
		To be discussed in Lab	Hardware Description Languages	Modeling with HDLs Verilog/VHDL
3,4	3	7-11	Reconfigurable Computing Device	FPGA Architecture, FPGA Fabrics Configuration - SRAM Based-FPGAs - Permanently Programmed FPGAs Programmable I/O, Circuit Design of FPGA Fabrics, Architecture of FPGA Fabrics, Case Studies (Xilinx/Altera/Microsemi/Actel).
5		12-15	Reconfigurable Computing Architecture	Fine - Grained and Course - Grained Reconfigurable Architecture, Case Studies.
6	4	16-18	Programming Reconfigurable Systems	Logic Design Process - Design - Integration - FPGA Design Flow Implementation Approaches - Run Time Reconfiguration (RTR) - Partial Reconfiguration (PR)
7,8		19-24	Mapping Designs to Reconfigurable Platform	Logic Implementation for FPGAs, Syntax-Directed Translation Logic Synthesis - Two-Level Logic Synthesis - Multi-Level Logic Synthesis LUT-Based Technology Mapping
9-12		25-35	High-Level Synthesis for Reconfigurable Devices (Behavioral Design)	Modeling - DFG, CFG Introduction to Binding, Scheduling and Allocation, Temporal Partitioning Algorithms - ASAP - ALAP
13		36-39	Temporal Placement and Routing	Offline and Online Temporal Placement Routing Cost, Routing-Conscious Placement
13,14		39-40	Online Communication	Communication at run-time between modules on

				the Reconfigurable Device
14		41-43	Reconfiguration Management	Multi-Context FPGAs, Introduction to Partial Reconfiguration, protecting the FPGA design from common threats, Design security concerns, Secure architecture in FPGAs and SoC FPGAs.

8. Evaluation Scheme

EC No.	Evaluation Component	Duration (min)	Marks (Weightage %)	Date and Time	Nature of Component
1	Mid-Semester Exam	90 Min	25%		Closed Book
2	Regular Labs/ Lab Assignments /Assignments	NA	20%	Regular/ To be announced	Open Book
3	Project	NA	20%	Regular/ To be announced	Open Book
4	Comprehensive Exam	3hrs	35%		Closed Book

9. Lab: This course has lab components using Xilinx Vivado suite. Final design should be implemented in Xilinx FPGAs.

10. Chamber Consultation Hour : Monday 4-5pm. Students can meet me at other times after requesting an appointment via email: meetha.shenoy@pilani.bits-pilani.ac.in

11. Notices: Notices regarding the course will be put up on the course web site

12. Makeup Policy : In general, Make-up will not be granted without prior permission from IC. If the student is unable to appear for the Mid-Semester Test/ Comprehensive Examination due to exigencies, the student must refer to the AUGSD stipulated procedure for applying for Make-up.

13. Note (if any): It shall be the responsibility of the individual student to be regular in attending lectures and the lab sessions as per the schedule announced in time table.

Instructor-in-charge:
Dr. Meetha V Shenoy
Course No. CS G553